

Scalability & Future Plan

Date	04 July 2026
Team ID	
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	1 Mark

Scalability & Future Plan

This document outlines the scalability and future enhancement plans for the Human Development Index (HDI) Prediction System, including advanced machine learning models, cloud deployment, secure user management, and support for larger datasets and users.

Current System Limitations

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	The application predicts only the Human Development Index (HDI) and does not include other development indices such as GII or MPI.	Limits the overall analysis of human development.	High
2	The application is currently deployed locally and cannot be accessed through the internet.	Limits accessibility for users outside the local system.	High

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
3	Prediction history is stored using Local Storage instead of a centralized database.	Limits long-term data storage, security, and multi-user access.	Medium

Scalability Plan

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports limited local users	Deploy the application on cloud platforms such as IBM Cloud, AWS, or Azure to support multiple concurrent users.
2	Data Storage	Local Storage is used for prediction history	Replace Local Storage with a relational database such as PostgreSQL, MySQL, or SQLite using SQLAlchemy.
3	Performance	Linear Regression model performs predictions locally	Improve prediction performance by integrating advanced Machine Learning models such as Random Forest and Gradient Boosting.
4	Security	Basic session-based authentication	Implement secure password encryption, HTTPS, and role-based user authentication for enhanced security.

Future Roadmap

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Integrate Gender Inequality Index (GII) and Multidimensional Poverty Index (MPI) prediction modules.	Within 2 Months	Provides a more comprehensive analysis of human development.
Phase 2	Implement advanced Machine Learning models and time-series forecasting for future HDI prediction.	Within 4 Months	Improves prediction accuracy and enables future trend analysis.
Phase 3	Cloud deployment, database integration, Docker support, and mobile-friendly interface.	Within 6 Months	Enhances scalability, accessibility, performance, and user experience.